

This is a draft offered in advance of the assessment period; there may be changes between now and then.

0.1 Marking Criteria

Reports will be marked according to the **University of Bristol Marking Criteria**. Below is a summary of how these criteria apply to this report.

Mark Range	Criteria
100–80	All plots completed and correct; Excellent presentation; Outstanding critical analysis and discussion of results; Evidence of study and originality well beyond what has been taught. Mathematics used to aid communication, referenced in insightfully, and free from typographical issues.
70–79	All plots completed and correct; Excellent presentation; Good critical analysis and discussion of results; Some evidence of study and originality beyond what has been taught. Mathematics used to aid communication, and free from typographical issues.
60–69	Almost all plots completed and correct; Good presentation; Some critical analysis and discussion of results. Mathematics used to aid communication.
50–59	Most plots completed and correct; Coherent presentation; Some discussion of results. Mathematical expressions are included, albeit tangentially.
40–49	Some plots completed and correct; Coherent presentation; Attempt at discussion of results. Some mathematical expressions present.
0–39	Insufficient plots completed and correct; Understandable presentation; Little or no discussion of results. Mathematical expressions are missing or contain mistakes that suggest misunderstanding.

Marks above ~ 60 require not only most plots completed and correct, but also critical analysis and discussion of the results. What does this mean? Computational neuroscience is a science. In science, we don't just compute numbers and make plots. Those numbers and plots are supposed to tell us something about the brain. It is important to think about the following issues when answering the written parts:

- Do the numbers/plots we've computed make sense?
- What do the numbers/plots tell us about the underlying system?

- Do the numbers/plots really capture what we want them to capture? (e.g. is FF/CV a “good” measure of variability?)

The remaining pages present further marking notes for students who are anxious about details, but may also be a bit of information overload. These notes are provided to clarify what we mean by words like “professionally”, “correct”, “coherent”, etc., since these terms may be vague or culturally relative.

0.1.1 Technical content

- Complete all numerical analyses and simulations correctly, and provide sufficient detail in your write-up to show the marker what you did, and convince them that your work is correct.
- Describe your approaches and algorithms clearly and with sufficient precision that others could reproduce them.
- Include selected, small code excerpts in the main text to clearly communicate your solutions.
- Clearly state your assumptions and approach on questions that are open-ended or have multiple interpretations. Explain why you used a particular assumption, or chose a specific method.
- Clearly and accurately contextualise all results in terms of the relevant neurophysiology. Demonstrating applicable neuroscience knowledge is required for full marks.
- Provide equations and describe terminology for the reader, even if equations and descriptions are already provided in this document. This can be brief, but should be present.
- Avoid stating incorrect facts about neuroscience or misusing technical terms.
- The marker cannot provide credit for text that they cannot understand. Ambiguous, confused, or awkward statements of technical concepts may come across as a lack of understanding.

0.1.2 Figures, diagrams, and tables

- Use figures where appropriate to communicate results. You may find adding figures helps the narrative, even where the coursework has not specifically instructed you to add one.
- Refer to figures and sub-figures appropriately in the written text.
- Export and embed figures in a vector graphics format (.svg, .ps, .eps, .pdf). Standard programming environments can export these formats. Points will be deducted for pixelated bitmaps or screenshots. You can edit vector graphics in the free, open-source program Inkscape.

- Use consistent scaling and DPI across all figures embedded in the document. A good way to achieve this is to measure the width of a text column in your LaTeX template, choose 2 figure sizes, and use these figure sizes with a consistent DPI in your computer code. This means that the version of the figure you see while exploring is closer to the final version, so you can notice things to adjust earlier.
- Use a limited number (usually 2-4) readable font sizes, and keep these consistent across figures.
- Ensure that all graphs and axes have precise and clear labels that are as self-explanatory as possible.
- Report units for physical quantities.
- Give each figure a clear title that tells the reader what argument it supports. Imperative sentences in active voice are preferred.
- Clearly describe the figure in the caption and communicate what the reader should take away from the figure. If there are sub-figures, do this for each sub-figure and refer to sub-figure labels in the caption appropriately.
- Export and embed figures in a high-quality vector format (e.g. SVG, PDF, PS). If vector formats are not possible, ensure that raster images have sufficient resolution so that the figure does not appear pixelated when zooming in to read details.
- Use readable colour pallets, and adjust plot size, shape, axes limits, and scale to clearly show what is being plotted.
- Less is more: Old manuscripts, from a time when making complex figures was difficult, and provide inspiration: Recall how Hodgkin and Huxley presented their action potential. You probably do not need to repeat all axis labels or legends when you are preparing a group of similar sub-plots. Also consider simplifying your plots.
- If you are using Matplotlib, the seaborn package may provide cleaner defaults. You may find you need to write a bit of code to “clean up” the default figure options of your chosen programming environment.

0.1.3 Write clearly and to a professional standard

- Write in complete sentences and organise your text into paragraphs.
- Use a clear narrative style for each specific question. You do not need to link the entire coursework into a single linear narrative, but your answers to each question should flow clearly.
- Use each (sub-)figure to support an argument in the main text, and include cross-references each (sub-)figure in the text.
- Incorporate and reference mathematical equations to make your explanations precise.

- Avoid overly long sentences.
- A brief, clear, direct style may help the marker understand your approach and recognise your expertise. A more lengthy style is also fine—provided the writing is clear.
- Avoid spelling errors or grammatical mistakes.
- Use clear and unambiguous language throughout.
- Define acronyms and abbreviations before using them.
- Avoid excessive passive voice: If you want to do this while avoiding first-person pronouns I/we, you can make the method, routine, or approach the agent in the sentence. For example, simulated revealed that... Inspection of the Fano factors showed that... etc.
- Avoid awkward or inverted clause structures. Excessive use of commas is sometimes a warning sign of this.
- Avoid wordiness. **G. Kim Blank's wordiness list provides useful substitutions.** Check whether replacing longer words with plainer English (e.g. replacing utilise with 'use', and novel with 'new') makes a given sentence sound more authoritative or clear.
- This coursework is designed to be self-contained. You are not required to cite additional sources. However, you may cite outside sources if you referred to them. Use a professional citation style with inline cross-references and a consistently formatted bibliography.

0.1.4 Avoid common issues in mathematical typesetting

- Font styles of variable names are typically contrasted in mathematics. For example, $X, X, x, \mathbf{x}, x, \mathcal{X}, x, \mathcal{X}, x$, and X may be *different* objects. Keep variable name styling consistent throughout the document and figures.
- Format text embedded in equations as text, as in

$$\dot{p}_{Na} = f(v, p_{Na}) \leftarrow \text{This the ODE} \quad (1)$$

and not

$$\dot{p}_{Na} = f(v, p_{Na}) \leftarrow \text{This the ODE}, \quad (2)$$

which typesets Na as if $N \cdot a$.

- Equations should read as part of grammatically correct sentences. This is true even for equations that are displayed on their own line. Notice how Equations (1) and (2) read as part of the text, complete with a comma after (2) to introduce the “which...” clause that follows.
- Take care to typeset operators like $\cos(\theta)$ as operators and not as $\cos(\theta)$, which sometimes looks too much like e.g. $c \cdot o \cdot s(\theta)$.

- Number equations if you wish to refer to them in the text, for example Equation (1) handled text embedded within the equation correctly.
- The most universally accepted formats for scientific notation are 3×10^{-4} or $3 \cdot 10^{-4}$. Some editors and reviewers may object to the format `3e-4` unless it is clear from context that `3e-4` is a valid floating-point literal in the programming language being used. That said, you *will not* lose marks for using the `3e-4` format anywhere in your document.
- Avoid calling the horizontal and vertical axes the x -axis and y -axis, respectively, unless you are actually plotting a variable named x on the horizontal axis or a variable named y on the vertical axis.